

3. PROJECT DESIGN PHASE

3.1 Introduction

The Project Design Phase defines the structure, components, workflow, data flow, and user interaction of the FitBuddy – AI Fitness Plan Generator. The main objective of this phase is to convert the requirements identified during the analysis phase into a clear and implementable system design. FitBuddy is designed as a web-based application that accepts user fitness information and uses an AI model to generate a personalized fitness and nutrition plan.

3.2 Design Objectives

The major objectives of the design phase are:

- To design a simple and user-friendly interface for entering fitness details.
- To organize the application into frontend, backend, database, and AI service components.
- To provide a clear communication flow between the user interface and the backend.
- To integrate the Gemini AI model for personalized plan generation.
- To store required user and generated-plan information in a structured database.
- To make the system modular so that individual components can be maintained or enhanced easily.

3.3 Overall System Architecture

FitBuddy follows a layered architecture. The frontend provides the user interface, while the backend handles requests, validation, database operations, and communication with the AI model. The database stores application data, and the Gemini model processes the fitness-related prompt prepared by the backend.

Layer	Main Responsibility
Presentation Layer	Collect user information and display the generated fitness plan.
Application Layer	Process requests, validate inputs, prepare prompts, and control application logic.
AI Layer	Generate personalized workout and nutrition recommendations using Gemini.
Data Layer	Store and retrieve user information and generated plan details using SQLite.

The architecture separates responsibilities between components, which helps improve maintainability, testing, and future expansion of the application.

3.4 Module Design

The system is divided into functional modules. Each module performs a specific task and communicates with the other modules through well-defined inputs and outputs.

A. User Input Module

This module collects information such as age, gender, fitness goal, activity level, available workout time, diet preference, and other relevant details. Input validation is performed before the data is sent to the backend.

B. Backend Processing Module

The backend receives the submitted information through API endpoints. It validates the request, prepares the required data structure, and controls the overall processing sequence.

C. AI Integration Module

The AI integration module prepares a suitable prompt from the user's information and sends it to the Gemini model. The generated response is received by the backend and formatted for presentation.

D. Workout Plan Module

This module handles the workout portion of the generated result. The plan can include exercises, repetitions or duration, rest guidance, and a weekly structure according to the information supplied by the user.

E. Nutrition Plan Module

This module handles the nutrition portion of the AI-generated response. It can organize meal suggestions and general nutrition guidance according to the user's stated preferences and goals.

F. Database Module

The database module manages persistent application data. SQLite is used as the database for the project because it is lightweight and suitable for a small academic web application.

3.5 Functional Flow

Step	Process
1	User opens the FitBuddy application.
2	User enters fitness and preference information.
3	Frontend sends the information to the backend API.
4	Backend validates and processes the information.
5	Backend prepares a structured prompt for Gemini.
6	Gemini generates the requested personalized plan.
7	Backend formats the response and returns it to the frontend.
8	Frontend displays the generated workout and nutrition plan.

3.6 Data Flow Design

The Data Flow Design explains how information moves through FitBuddy. The user provides input through the frontend. The backend receives and validates the input, then communicates with the AI service. The generated response is returned to the application and displayed to the user.

Level 0 – Context Flow

At the highest level, FitBuddy can be represented as a single system interacting with two external entities: the User and the Gemini AI Service. The user supplies fitness requirements and receives a generated plan. The application sends structured prompts to the AI service and receives generated content.

Level 1 – Major Data Flow

- User → Frontend: Fitness details and preferences.
- Frontend → Backend API: Validated form data.
- Backend → Database: User and plan information when persistence is required.
- Backend → Gemini: Structured AI prompt.
- Gemini → Backend: Generated fitness-plan response.
- Backend → Frontend: Processed plan.
- Frontend → User: Final workout and nutrition information.

3.7 Database Design

The database design provides a structured method for storing information required by the application. A simple relational structure can be used for the academic implementation.

Field / Entity	Description
User ID	Unique identifier for a user record.
Age	Age supplied by the user.
Fitness Goal	Goal selected by the user, such as general fitness.
Activity Level	User's stated activity level.
Diet Preference	Diet preference entered by the user.
Workout Preference	Preferred workout details or available time.
Generated Plan	AI-generated workout and nutrition content.
Created Date	Date/time associated with the generated record.

3.8 Input and Output Design

Input design focuses on collecting only the information required for generating a useful response. Output design focuses on presenting the result in a readable structure, separating workout guidance from nutrition guidance and making the result easy to understand.

3.9 User Interface Design

The FitBuddy interface is designed with simplicity and ease of use as the main considerations. The user should be able to understand what information is required without technical knowledge.

Main Interface Components

1. Home / Introduction Section – Briefly explains the purpose of FitBuddy and guides the user to begin.
2. Fitness Details Form – Collects age, goal, activity level, workout preference, and related information.
3. Nutrition Preference Section – Collects dietary preferences required for generating general meal guidance.
4. Generate Plan Button – Sends the completed information to the backend for processing.
5. Result Section – Displays the AI-generated workout and nutrition plan.
6. Error / Validation Messages – Informs the user when required information is missing or invalid.

3.10 API Design

The backend exposes API endpoints that connect the frontend with application logic. The API design keeps the frontend independent from internal processing and makes the system easier to extend.

API Function	Method	Purpose
Generate Plan	POST	Accepts user fitness details and requests an AI-generated plan.
Get Plan	GET	Retrieves an existing generated plan when stored.
Health Check	GET	Checks whether the backend application is running.

3.11 Security and Validation Design

Basic validation is included to reduce invalid requests and improve reliability. Required fields should be checked before processing. API credentials such as the Gemini API key should be stored securely in environment configuration rather than directly exposed in frontend code. User inputs should also be handled carefully to avoid malformed requests and unexpected application behavior.

3.12 Error Handling Design

The system should handle common errors such as missing form values, invalid input, unavailable AI service, database errors, and unexpected server failures. Instead of exposing technical error details to users, the frontend should display a simple and understandable message.

3.13 Technology Design

The project design uses technologies that support rapid development and easy integration. The selected components are suitable for an academic AI-based web application.

Technology	Role in the Project
HTML / CSS / JavaScript	Frontend structure, styling, and browser-side interaction.
Python	Main backend programming language.
FastAPI	Backend framework for API creation and request handling.
SQLite	Lightweight database for storing application data.
Gemini AI	Generates personalized workout and nutrition plan content.
VS Code	Development environment used to create and manage the project.

3.14 Design Considerations

- Usability: The interface should be simple enough for first-time users.
- Modularity: Frontend, backend, database, and AI integration are separated logically.
- Maintainability: Clear modules and API endpoints make future modifications easier.
- Scalability: The architecture can be extended with additional features and database tables.
- Reliability: Validation and error handling reduce common runtime problems.
- Security: API credentials and sensitive configuration should not be exposed in client-side code.

3.15 Summary

The Project Design Phase establishes the technical blueprint of the FitBuddy – AI Fitness Plan Generator. The design defines the system architecture, modules, data flow, database structure, interface, API communication, validation, security considerations, and technology stack. This design provides a clear foundation for the implementation phase, where the planned components are developed and integrated into a working application.

3.16 Expected Design Outcome

The expected outcome of the design is a structured and user-friendly AI fitness planning system in which users can provide their requirements and receive a clearly organized, personalized fitness-plan response through the web application.